

Geographic independence and phylogenetic diversity of red shiner introductions

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Abstract Identifying areas at risk of invasion can be difficult when the distribution of a non-native species encompasses geographically disjunct regions. Understanding genealogical relationships among native and non-native populations can clarify the origins of fragmented distributions, which in turn can clarify how fast and far a non-native species may spread. We evaluated genetic variation across the native and invasive ranges of red shiner (*Cyprinella lutrensis*), a minnow known to displace and

hybridize with native species, to reconstruct invasion pathways across the United States (USA). Examination of mitochondrial cytochrome-*b* variation found that native range populations of red shiner fall into four highly divergent lineages that likely warrant species recognition. Introduced red shiner populations in the eastern and western USA are derived from only two of these lineages. Western USA populations originate from the mid-western and western genetic lineages, whereas eastern introductions derive only from the mid-western lineage. Western USA invasive populations exhibit fewer, but more diverse haplotypes compared to eastern USA invasive populations. We also recovered an undescribed, divergent lineage of *Cyprinella* that has been cryptically introduced into the western USA, which raises the possibility that hybridization has proceeded following secondary contact between previously allopatric lineages. Approximate Bayesian Computation modeling suggests that the disjunct distribution of red shiner across North America is an agglomeration of independent regional invasions with distinct origins, rather than stepwise advance of an invasion front or secondary introductions across regions. Thus localized control may be effective in managing non-native red shiner, including further spread to areas of conservation concern.

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Introduction

Biological invasions are a global threat to aquatic biodiversity (Wilson 1992; Sala et al. 2000). Human transport and introduction of non-native species has contributed to

widespread restructuring and biotic homogenization of aquatic communities (Moyle and Leidy 1992; Allan and Flecker 1993; Olden et al. 2004). Of particular concern are non-native freshwater fishes, which have higher establishment rates than other taxa (Jeschke and Strayer 2006), because introductions can result in the displacement and extirpation of native biota through predation (Witte et al. 1992), competition (Callaway and Aschehoug 2000) and hybridization (Olden et al. 2004). Non-native fish introductions can devastate populations of at-risk native species and irrevocably restructure entire assemblages. For example, the introduction of non-native trout to historically fishless, high elevation lakes precipitated the decline of threatened amphibians in the Sierra Nevada mountains in the western United States (USA) and the introduction of predatory Nile perch (*Lates nilotica*) has resulted in the loss of nearly 200 endemic cichlids in Lake Victoria (Kaufman 1992; Knapp and Matthews 2000).

Freshwater aquatic invasions, including the advance of predatory and highly aggressive fish species, are often a result of intentional or accidental introductions related to aquaculture and angling (Lever 1996; Moyle and Light 1996). Aquaculture and angling have been identified as primary sources for fish introductions worldwide, with 50 % of cultivated freshwater fish species having become established in the wild (Casal 2006). For example, the golden shiner (*Notemigonus crysoleucas*), which is a widely used bait fish, is now widespread across western North America (Dill and Cordone 1997; Pflieger 1997). Similarly, three species of Asian carp (*Hypophthalmichthys nobilis*, *H. molitrix*, *Mylopharyngodon piceus*), which were accidentally released from aquaculture facilities, are progressing rapidly across the Mississippi River Basin (Chick and Pegg 2001). Though invasions can result in a cohesive distribution that reflects expansive spread of a unified invasion front, like the progression of Asian carp in the Mississippi River drainage, non-native freshwater fishes often exhibit patchy distributions (Townsend and Crowl 1991; Leprieur et al. 2006). Despite the prevalence of patchy distributions, the origins and interactions that occur across disjunct populations are not well understood, which can complicate efforts to determine how fast and far non-native freshwater fish may spread.

Several processes may give rise to fragmented distributions of invasive species. Multiple independent introductions from distinct sources within the native range can result in disjunct static populations with little to no interaction, though post-introduction interactions with native species can trigger unification through expansive spread. For instance, the sheepshead minnow (*Cyprinodon variegatus*) was introduced to the Pecos River drainage in western Texas and New Mexico via bait bucket releases between 1968 and 2000. Sheepshead minnow eventually

hybridized with the imperiled Pecos pupfish (*Cyprinodon pecosensis*), resulting in the spread of hybrids across more than half of the range of the Pecos pupfish in less than 5 years (Echelle and Connor 1989; Childs et al. 1996; Rosenfield et al. 2004). Disjunct populations can also arise from stochastic secondary spread following a single introduction event. For example, the spread of invasive European green crabs (*Carcinus maenas*) across the northeastern Pacific coast resulted from stochastic transport by climate-mediated currents following a single introduction from a small founding population derived from the Atlantic coast of North America (Darling et al. 2008; Tepolt et al. 2009). The northeastern Pacific coast range of invasive *C. maenas* ballooned in less than 10 years with the establishment of disjunct populations isolated across distances as large as 1000 km (Tepolt et al. 2009). Similarly, the progression of Asian topmouth gudgeon (*Pseudorasbora parva*)—which preys on eggs and transmits the parasite *Sphaerothecum destruens* to native fishes—across 32 European countries is likely the result of a single introduction into Romania or Hungary followed by stochastic secondary spread (Gozlan et al. 2010; Simon et al. 2011).

Reconstructing genealogical relationships among native and non-native populations can clarify the origins and nature of fragmented distributions (Estoup and Guillemaud 2010). For example, Blum et al. (2007) were able to validate historical evidence that smooth cordgrass (*Spartina alterniflora*) was introduced into Willapa Bay (Washington) from multiple source populations, and found evidence suggesting *S. alterniflora* secondarily spread to a nearby embayment on the Pacific coast. Similarly, studies of the invasive European green crab in the Canadian Maritimes and New England coast revealed that geographic differences in genetic diversity were best explained by multiple introduction events spanning a period of several decades (Roman 2006). Subsequent work (Blakeslee et al. 2010) determined that introduced populations in Newfoundland originated from Nova Scotia via counter-current spread, rather than directly from Europe. Finally, studies of the western mosquito fish (*Gambusia affinis*) across the North Island of New Zealand demonstrated that the establishment and spread of *G. affinis* reflect a combination of localized dispersal and human-assisted colonization (Purcell et al. 2012; Purcell and Stockwell 2015).

Understanding the relative contributions of different processes that give rise to disjunct distributions can help identify areas at risk of invasion as well as improve the selection and impact of management approaches. In this study, we evaluated genetic variation across the native and invasive ranges of red shiner (*C. lutrensis*) to infer relationships among invasive populations and to reconstruct pathways of spread across the eastern and western USA. Non-native red shiner populations are representative of the

legacy of intentional and accidental introductions from aquaculture and angling in North America. Historical collection records suggest that red shiner were introduced to the western USA at least 20 years before being introduced to the eastern USA (Table 1). However, it is not clear whether the emergence of non-native red shiner populations across the USA is (1) an outcome of multiple, independent introductions from the native range; or (2) stepwise advance through successive secondary introductions from one part of the invasive range to another. We evaluated both possibilities by first examining mtDNA haplotype variation across the native range to characterize sources that could have contributed to introduced populations. We then examined mtDNA haplotype variability among representative samples from the invasive range to infer the source and number of introductions across the eastern and western USA. Finally, comparisons of native and invaded regions enabled us to infer the sequence and pathways of invasion, which in turn offered novel perspectives on risk and prevention of further spread to areas of conservation concern.

Materials and methods

Study species

Red shiner now occur in 11 states beyond native regions in North America (Fuller et al. 1999). With a native range encompassing Great Plains and Central Lowland tributaries of the Mississippi River and western Coastal Plain drainages of the Rio Grande River, red shiner range from

California to North Carolina. The species thrives under harsh conditions (e.g., low flow, high turbidity, poor water quality) and aggressively colonizes severely degraded habitats (Cross and Cavin 1971; Matthews 1985; Matthews and Hill 1977, 1979). For example, introduced red shiner have become the most abundant species in degraded, urban streams in metropolitan Atlanta, Georgia (Devivo and Freeman 1995). Due to high fecundity and aggressive behavior, introduced populations can spread rapidly, often displacing native species and hybridizing with native congeners (Hubbs and Strawn 1956; Minckley and Deacon 1968; Page and Smith 1970; Greger and Deacon 1988; Larimore and Bayley 1996; Moyle 2002; Walters et al. 2008; Blum et al. 2010; Ward et al. 2012; Glotzbecker et al. 2015). Dill and Cordone (1997) referred to the red shiner as the second greatest threat to the welfare of indigenous south-western fishes (after the mosquito fish, *Gambusia affinis*). Red shiner populations in the Moapa River and Virgin River (Nevada) have been implicated in the regional decline of native fish including spikedace (*Meda fulgida*), woundfin (*Plagopterus argentissimus*), and Virgin River chub (*Gila seminuda*; Moyle 1976; Deacon 1988; U.S. Fish and Wildlife Service 1990, 1995). The decline of redbreasted shiner (*Richardsonius balteatus*) also coincided with the rise of red shiner in the Green River outside of Dinosaur National Monument (Utah; Holden and Stalnaker 1975). Additionally, over 40 % of congeners in the southeastern USA, widely considered to be a global hotspot of aquatic biodiversity, are known to hybridize with introduced red shiner in the wild (*C. analostana*, *C. camura*, *C. callitaenia*, *C. spiloptera*, *C. venusta cer-costigma*, *C. v. stigmatura*, *C. v. venusta*, and *C. whipplei*; Hubbs and Strawn 1956; Page and Smith 1970; Wallace and Ramsey 1982; Johnson 1999; W.C. Starnes personal communication) or under laboratory conditions (*C. caerulea*, Burkhead unpublished data).

Time series collection records indicate that red shiner were first introduced to the western USA in the Lower Colorado River in the 1950s and then later introduced to the eastern USA in the Lower Yadkin River and Upper Coosa River in the early 1970s (Fuller et al. 1999; Walters et al. 2008; Table 1). Given the distance between these geographic regions, it is likely that non-native populations in the western and eastern USA are a result of multiple independent introductions from the native range. Survey records also suggest that the progression of red shiner invasions within drainage basins may be the result of multiple independent introduction events (Walters et al. 2008). However, it is possible that non-native populations originated from secondary introductions (e.g., Darling et al. 2008) as a consequence of human mediated bait bucket or aquarium releases deriving from the invaded range (Jennings and Saiki 1990; Walters et al. 2008).

Table 1 Collection records of first occurrences of non-native red shiner

State	Year	Drainage or river
California	1950	Lower Colorado and Whitewater River
Arizona	1953	Lower Colorado
Illinois	1958	Lake Michigan
Utah	1962	Upper Colorado
Nevada	1967	Lower Colorado and Lower Virgin River
Colorado	1969	Lower Yampa
Alabama	1970	Chattahoochee River
North Carolina	1974	Lower Yadkin
New Mexico	1980	Gila River
Wyoming	1982	Great Divide-Upper Green River
South Carolina	1986	Lower Pee Dee
Virginia	1986	Roanoke
Georgia	1992	South Atlantic Gulf

Predictive models and inventory records suggest that red shiner could spread throughout the conterminous USA (Poulos et al. 2012; Poulos and Chernoff 2014) as a consequence of rapid population growth and aggressive colonization (Hubbs and Lagler 1964; Minckley and Deacon 1968; Minckley 1973). Based on the premise that red shiner have broad environmental tolerances, forecasting models indicate that red shiner can potentially invade habitats with mean minimum water temperatures above freezing and mean maximum summer water temperatures up to 35 °C (Matthews and Hill 1977; Poulos et al. 2012). According to some climate change projections, the range of native and non-native red shiner populations could increase by 42 %. Projected expansions into the western USA, southeastern USA, and much of Canada (Poulos and Chernoff 2014) could precipitate widespread declines in the abundance and distribution of other stream fishes (Gido et al. 1999; Marsh-Matthews and Matthews 2000; Douglas et al. 1994).

Native and invasive range collections

Between 2004 and 2010, specimens were obtained from 43 rivers (34 in the native range, 9 in the invasive range) across the USA. Specimens of native red shiner were collected from the Mississippi River between Louisiana and Iowa. Specimens also were sampled from basins in Texas, New Mexico, Oklahoma, Kansas, Nebraska, Illinois, and Arkansas (Supplementary Table 1). Specimens of non-native red shiner were sampled from western USA populations in California, Nevada, and Arizona (Supplementary Table 1). Non-native red shiner from the eastern USA were collected in Alabama, Georgia, and North Carolina (Supplementary Table 1). Though we obtained samples representing all known clades and lineages of red shiner within the native range, we were unable to obtain samples from non-native populations in the northern reaches of the Colorado River and Green River, as well as small populations recorded in Virginia and Massachusetts. We were also unable to obtain red shiner from aquaculture stock, as their use is now largely prohibited in the bait and ornamental fish industry. This prevented us from drawing potentially valuable comparisons between introduced, native, and aquaculture populations. On average, 10 individuals (SD = 8.87) were sampled at each site via seine netting (Supplementary Table 1). Once collected, whole fish were immediately preserved in 95 % ethanol for genetic analysis.

DNA extraction and mtDNA sequencing

Genomic DNA was extracted from approximately 0.05 g of preserved fin tissue from each specimen using DNeasy

kits (Qiagen, Inc., Valencia, CA). Approximately 10–50 ng of DNA was used as template for 15 µl polymerase chain reaction (PCR) mixtures that also included 2.5 mM MgCl₂, 2.5 mM each dNTP, 0.5 units *Taq* DNA polymerase (Invitrogen, Carlsbad, CA), 0.5 µM each of a pair of oligonucleotide primers and 1X PCR buffer (Invitrogen, Carlsbad, CA). The complete cytochrome *b* (*cyt b*) gene (1140 bp) was amplified with primers GLU and THR as described in Schmidt et al. (1998) under a thermal regime of 35 cycles of 94 °C for 30 s, 49 °C for 30 s, and 72 °C for 90 s, followed by a final extension stage at 72 °C for 5 min with a MJ Dyad thermocycler (MJ Research, Inc., Waltham, MA). PCR amplicons were purified with ExoSAP-IT (Affymetrix, Santa Clara, CA) following manufacturer protocols and served as templates in sequencing reactions utilizing ABI BigDye Terminator v.3.1 sequencing kits (Applied Biosystems, Inc., Foster City, CA). The sequencing reactions were run on an ABI 3730xl automated sequencer (Applied Biosystems, Inc., Foster City, CA). Raw sequence files were edited, assembled and aligned with Sequencher 5.0 (Gene Codes Corp., Ann Arbor, MI) in preparation for analysis. All sequences were translated into amino acids to check for stop codons, sequencing errors and misalignments. No indels were detected in any of the sequences. All newly obtained sequences were deposited in GenBank under accession numbers KR004184–KR004610. Separate accession numbers for previously acquired sequence data are provided in Supplementary Table 2.

Genetic data analysis

Phylogenetic relationships among haplotypes recovered in the native and invaded range were inferred using Maximum Likelihood (ML) analysis with the package phangorn (Schliep 2011) in R v.2.15.1 (R Development Core Team 2009). ML analyses were implemented using the TN93+I+Γ substitution model, which was selected according to log-likelihood values and Akaike information criterion (AIC) values in model tests run with PhyML v.3.0 (Guindon et al. 2010) and ape in R (Paradis et al. 2004). Node support was assessed from 500 bootstrap replicates. ML trees were rooted using *cyt b* sequences from *Nocomis biguttatus*, *N. micropogon*, *N. biguttatus* and *N. asper* (Accession Nos. GQ275147, GQ275148, GQ275149 and GQ275150; Schönhuth and Mayden 2010). In addition to including *cyt b* sequence data from other native range populations examined in Broughton et al. (2011) and Osborne et al. (2015), data from 33 recognized and proposed species of *Cyprinella* as well as representative species from nine genera of North American cyprinids were included in the analyses (Supplementary Table 2). Haplotype networks also were constructed for each of the major

lineages containing both native and non-native populations using the median-joining algorithm (Bandelt et al. 1999) in Network v.4.612 (Fluxus Technology Ltd., <http://www.fluxus-engineering.com>) to further explore genealogical relationships as well as haplotype distributions and frequencies in the native and invasive ranges of red shiner.

Estimates of genetic diversity were calculated for every sample site as well as for the major lineages recovered in phylogenetic analyses. DnaSP v.5.10.1 (Librado and Rozas 2009) was used to calculate the number of haplotypes and to estimate mitochondrial haplotype diversity (H_d), nucleotide diversity (π), the number of polymorphic sites (S), the average number of nucleotide differences between sequences (k), and the Watterson's estimator of diversity (θ_S) which is a measure of the effective population size for neutral mutations, for each river and major lineage. We conducted AMOVAs in Arlequin v.3.5.1.2 (Excoffier and Lischer 2010) to characterize patterns of hierarchical (collection site, river, and drainage) genetic variation within and between the native and invasive ranges. Genetic variation in the native range was tested independently for each of the major lineages to account for the unresolved phylogeny of *Cyprinella*, as some lineages are thought to include multiple species (Schönhuth and Mayden 2010). Comparisons between the non-native ranges included all samples. Additionally, pairwise estimates of genetic differentiation (Φ_{ST}) were calculated between rivers using Arlequin v.3.5.1.2.

Testing models of invasion history

Competing scenarios of introductions were evaluated using Approximate Bayesian Computation (ABC) with the program DIYABC v.2.0 (Cornuet et al. 2014). Initially, we tested hypotheses describing the general pattern of invasion from the native into the disjunct invasive distributions in the eastern and western USA. We first examined five competing scenarios of colonization following collection records and records of introductions and the observed phylogeographic pattern. Consequently, only the major lineages occurring in the native and non-native ranges were included in the analyses (i.e., mid-western and western lineages; Supplementary Fig. 1). The first two scenarios assumed a single introduction of *C. lutrensis* from the native range into the western invasive range, and subsequent secondary introduction from the western to the eastern USA. Scenario 1 considers all non-native populations as derived from populations of the mid-western lineage, while Scenario 2 considers an ancestral population of mixed mid-western and western lineage individuals as the source of non-native populations (i.e., an admixed aquarium trade or bait fish stock). Two scenarios were proposed to assess the hypothesis of an independent origin of the

eastern and western invasive ranges. Scenario 3 assumes that the eastern invasive populations are derived from a single introduction of the native mid-western lineage, whereas the western invasive populations are derived from an introduction of the mid-western lineage and a second introduction of individuals from the native western lineage. Scenario 4, on the other hand, proposes that western non-native populations were initially founded by individuals from the western lineage and from a subsequent invasion of the mid-western lineage. As in the previous scenario, non-native eastern USA populations would have been founded exclusively from a mid-western lineage. Finally, Scenario 5 proposed the existence of an ancestral admixed population (i.e., an aquaculture fish stock) constituted by individuals from the mid-western and western lineages, which was the source for a single invasion of the western non-native range. In this scenario, the invaded range in the eastern USA would have been founded exclusively by individuals from the mid-western lineage (Supplementary Fig. 1).

Based on the results from testing competing scenarios of invasion history across the USA, we tested hypotheses of the invasion pathway within the eastern and western invasive ranges, respectively. We assessed whether the presence of red shiner within each region is a consequence of population expansion derived from one introduction or whether it reflects independent introductions at different times corresponding to historical records (Supplementary Fig. 1). For the western region, Scenario W1 assumes an initial introduction into the San Joaquin River, California in 1950 (i.e., our sampling site with the earliest record) and subsequent expansion in chronological order from a single introduction to other basins in the region: San Joaquin River, California in 1950; Virgin River, Nevada in 1967; Virgin River, Arizona in 1967; Gila River, Arizona in 1991 (Nico et al. 2015). Scenario W2, on the other hand, assumes multiple independent introductions from the native range into each sampled area within the western region following the same chronological order. Similarly, for the eastern region, Scenario E1 proposes that the initial colonization of Terrapin Creek, Alabama in 1992 (i.e., our sampling site with the earliest record) resulted in subsequent expansion according to chronological records (Terrapin Creek, Alabama in 1992; Dead River, Alabama in 1992; Buffalo Creek, North Carolina in 1994; Peach Tree Creek, Georgia in 1995; Conasauga River, Georgia in 2000; Rocky River, North Carolina in 2004; Nico et al. 2015), whereas Scenario E2 assumes that eastern populations are derived from multiple independent introductions from the native range following the same chronological order (Supplementary Fig. 1).

According to the historical records available, the prior distribution of demographic parameters considered all

introduction events to have occurred over the last 100 years (Supplementary Table 3). Also, we assumed that the effective population size was the same for all non-native populations and used a uniform distribution between 10 and 10,000 individuals, except for the native and ancestral populations which were bounded between 100 and 1,000,000 individuals. A reduction in effective population size was modeled in all introduction events from the native to the non-native populations (Supplementary Table 3).

Genetic variation within and between populations was characterized using a set of summary statistics, including for each population: number of haplotypes, number of segregating sites, mean of pairwise differences, variance of pairwise differences, Tajima's *D* and private segregating sites; and for each population pair: number of haplotypes, number of segregating sites, mean of pairwise differences and pairwise F_{ST} values. For the comprehensive and each regional analysis, we used 2×10^6 simulated datasets per scenario to build reference tables. To assess whether the model (scenario and parameter prior definitions) was on target (i.e. to ensure that at least one or more combinations of scenarios and priors could produce simulated data sets that parallel observed data to perform model comparisons), we performed a pre-evaluation step by evaluating all scenario and prior combinations in a Principal Component Analysis (PCA). The PCA was based on the space of parameter values and summary statistics on 100,000 simulated datasets from the different scenarios, and added the observed data on each plane of the analysis.

After verifying model suitability, we then compared the fit of the competing scenarios by calculating posterior probabilities using 1 % of the simulated datasets to estimate the relative posterior probability of each scenario with a logistic regression over a linear discriminant analysis of the summary statistics (Estoup et al. 2012). Confidence in the model of choice was assessed by calculating Type I (the probability of choosing a scenario when it is false) and Type II errors (the probability of rejecting a scenario when it is true) for each scenario.

Results

Haplotype diversity and distributions

We examined mitochondrial *cyt b* sequence data from 480 individuals from 46 locations, with data derived from 352 specimens sampled in the native range and 128 specimens collected in invaded regions. Though the majority of these data encompassed the full *cyt b* region, we constrained comparisons to a subsection of the gene in order to incorporate existing mtDNA sequence data from geographically

important regions. Thus, we reduced our sequence data set to an alignment targeting the first 574 bp fragment of the 5' portion of the *cyt b* gene for subsequent analysis. Analyses of this fragment resulted in the same phylogenetic relationships recovered in previous studies (Schönhuth et al. 2006, 2008; Schönhuth and Mayden 2010) using the complete *cyt b* gene. The resulting *cyt b* dataset yielded 132 haplotypes, with 103 haplotypes found in the native range and 29 recovered in the invasive range, which is comparable to levels of variation found in prior work on the genus *Cyprinella* (Richardson and Gold 1995; Schmidt et al. 1998; Schönhuth and Mayden 2010) and *C. lutrensis* (Broughton et al. 2011; Osborne et al. 2015).

The phylogenetic hypothesis recovered from ML analysis of mtDNA sequence variation indicated that our samples constitute four highly supported clades (i.e., ≥ 98 % bootstrap support; Fig. 1). Although the relationships among the lineages are not well resolved, it is clear the four lineages do not form a monophyletic group and that each lineage corresponds to a distinct geographic region: the mid-west, which includes rivers in the Mississippi River drainage; the south-west, which includes rivers in the Rio Grande River drainage; the west, which includes rivers from the Gulf Coast, Rio Grande and Mississippi River drainage; and the Brazos, which includes rivers draining to the Gulf of Mexico largely within Texas. The range size of the four lineages is variable, with the mid-western lineage being the largest and most widespread, followed by the western, south-western, and Brazos. Genetic divergence among lineages ranges between 8.23 ± 1.05 % (uncorrected p-distance between mid-west and Brazos lineages) and 12.36 ± 1.27 % (between the west and south-west lineages). In addition, genetic diversity varies across the lineages, with the mid-western lineage encompassing 78 haplotypes, the south-western clade encompassing 23 haplotypes, the western clade encompassing 16 haplotypes, and the Brazos clade encompassing 13 haplotypes (Supplementary Table 1). The most widespread and abundant haplotypes were members of the mid-western lineage.

Examination of introduced populations recovered the presence of two clades in the western USA, and one clade in the eastern USA. Only the mid-western and western lineages included haplotypes recovered in invaded regions (Figs. 1, 2). The mid-western lineage included haplotypes recovered in invasive populations in the eastern and western USA, including: Dead River (AL), Terrapin Creek (AL), Virgin River (AZ), Conasauga River (GA), Peachtree Creek (GA), Buffalo Creek (NC), and Rocky River (NC; Figs. 1, 2). The western clade included haplotypes only recovered in invasive populations sampled in California and haplotypes shared between California and native range populations (Figs. 1, 2).

The figure displays a phylogenetic tree of *Crotaphytus* species, rooted with an OUTGROUP. The tree is color-coded by geographic region: MIDWEST (pink), BRAZOS (grey), WEST (blue), SOUTHWEST (green), and RED RIVER (orange). Bootstrap values are shown at the nodes. Species names and their distributions are listed to the right of the tree, accompanied by maps of the United States showing the specific drainage areas.

- MIDWEST:** *C. lutrensis* (Mississippi - R. Grande). Includes haplotypes Hap. 121, Hap. 36, Hap. 3, STL660.01 (GQ275188), STL767.01 (GQ275186), Hap. 70, Hap. 59, Hap. 23, Hap. 42, CL702 (GQ275185), UAIC11564.03 (GQ275184), Hap. 73, Hap. 2, Hap. 53, Hap. 19, Hap. 14, Hap. 37, Hap. 34, Hap. 86, Hap. 58, Hap. 88, Hap. 50, Hap. 48, Hap. 49, Hap. 68, Hap. 53, Hap. 72, Hap. 12, Hap. 63, Hap. 22, Hap. 44, Hap. 52, Hap. 43, Hap. 55, Hap. 83, Hap. 66, Hap. 87, Hap. 64, Hap. 61, Hap. 39, Hap. 4, Hap. 18, Hap. 62, Hap. 57, Hap. 51, Hap. 10, Hap. 20, Hap. 15, Hap. 16, Hap. 17, Hap. 9, Hap. 13, Hap. 21, Hap. 74, Hap. 40, Hap. 54, Hap. 38, Hap. 80, Hap. 35, Hap. 32, Hap. 120, Hap. 6, Hap. 85, Hap. 84, Hap. 11, Hap. 1, Hap. 41, Hap. 46, Hap. 45, Hap. 78, Hap. 76, Hap. 67, Hap. 75, Hap. 77, Hap. 79.
- BRAZOS:** *C. lutrensis* (Mississippi - R. Grande). Includes haplotypes Hap. 47, Hap. 100, Hap. 98, Hap. 71, Hap. 93, Hap. 91, Hap. 94, Hap. 97, Hap. 115, Hap. 96, Hap. 101, Hap. 92, UAIC112434.01 (GQ275187), Hap. 95, Hap. 99, Hap. 89.
- formosa:** *C. formosa*. Includes haplotypes Hap. 132, Hap. 130, Hap. 129, Hap. 131, Hap. 128, UAIC14270.02 (GQ275199), CFI01 (GQ275186).
- forlonensis:** *C. forlonensis* (Pánuco - Tamesi Drainage). Includes haplotypes UAIC7902 (DQ324098), UAIC9155.02 (DQ324096), MNCN2655 (DQ324094), BFK9065 (GQ275178).
- lepida:** *C. lepida* (Frio River). Includes haplotypes UAIC8348.03 (GQ275177), Hap. 134, Hap. 133, Hap. 124, Hap. 125, Hap. 135.
- lutrensis WEST:** *C. lutrensis* WEST. Includes haplotypes Hap. 122, Hap. 8, Hap. 29, Hap. 28, Hap. 56, Hap. 25, Hap. 116, Hap. 123, Hap. 26, Hap. 24, Hap. 118, Hap. 51, Hap. 27, Hap. 117.
- whipplei:** *C. whipplei* (Red River). Includes haplotypes UAIC7963.06 (GQ275230), STL281.02 (GQ275231).
- suavis:** *C. suavis* (Gulf Slope drainages). Includes haplotypes Hap. 82, UAIC9352.02 (DQ324095), CLBC2 (GQ275189), Hap. 103, MNCN2433 (GQ275190), Hap. 105, Hap. 112.
- cf. lepida:** *C. sp. 'cf. lepida'* (Nueces River). Includes haplotypes Hap. 137, Hap. 136, STL13131.02a (GQ275176), STL13131.02a (GQ275175), Hap. 113, Hap. 114.
- garmani:** *C. garmani*. Includes haplotypes CPUM1468 (EU082523), CPUM1449 (EU082522), C. garmani Hap. 127, C. garmani Hap. 126, C. garmani (DQ324102), UAIC13899.01 (GQ275194), Hap. 81, Hap. 104, Hap. 69, DAN0641 (GQ275191), Hap. 90, DAN0639 (GQ275192), RLM0701 (GQ275193), Hap. 111, Hap. 109, Hap. 108, Hap. 102, Hap. 107.
- lutrensis Pecos River:** *C. lutrensis* (Pecos River). Includes haplotypes Hap. 60, CL0637 (GQ275183), Hap. 119.

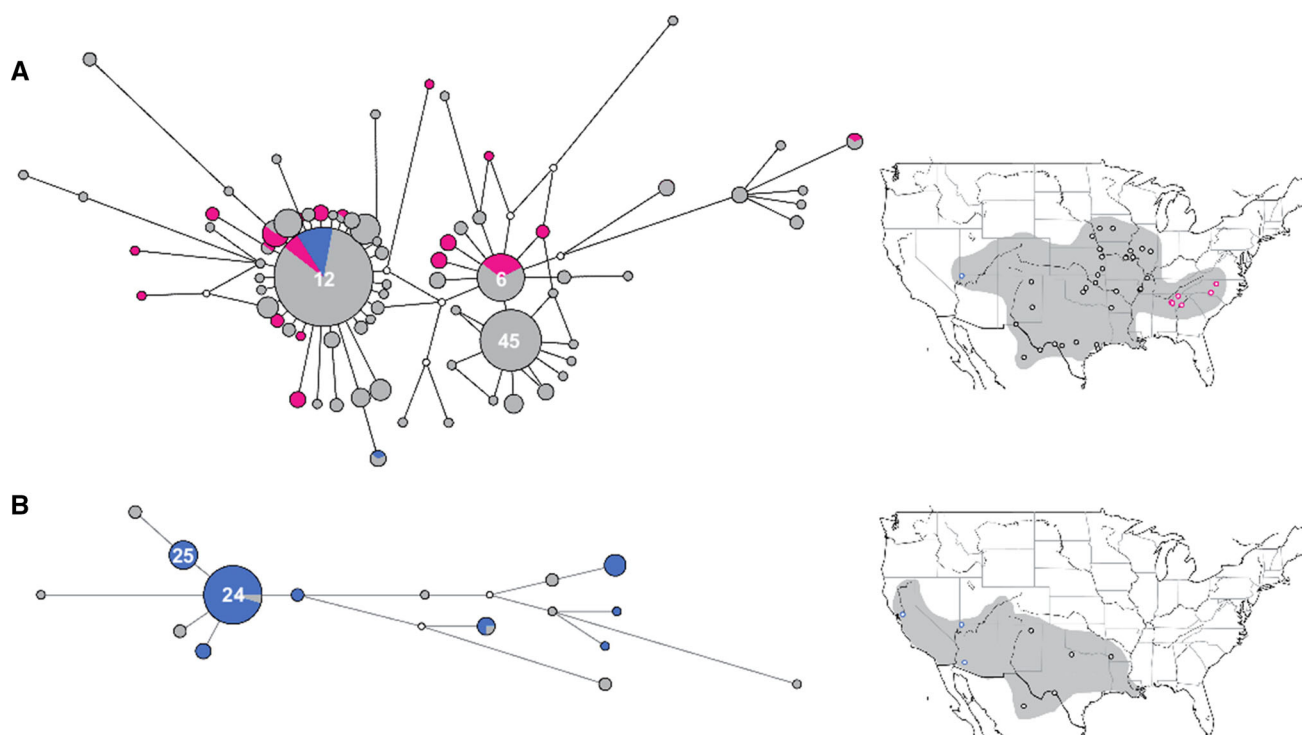


Fig. 2 Median-joining networks for those lineages of *Cyprinella lutrensis* occurring in native and non-native populations and their distribution ranges. **a** Mid-western lineage, **b** western lineage. *Blue outlined circles* haplotypes occurring in the western USA invasive population; *black outlined circles* haplotypes occurring in native

range populations; *pink outlined circles* haplotypes occurring in eastern USA invasive populations. The most common haplotypes are labeled by number (see main text). Internode length is proportional to the number of substitutions between haplotypes, and node size indicates relative haplotype abundance. (Color figure online)

The haplotypes recovered in invasive populations were not prevalent within the mid-western lineage, whereas the western clade haplotypes recovered in invasive populations dominated the clade (Figs. 1, 2). Of these, six haplotypes were shared between native and invasive ranges in the mid-western clade, and only two haplotypes were shared between native and invasive ranges in the western clade. A single haplotype (haplotype #12) was recovered in approximately one quarter of all samples ($n = 109$), and was found in eastern and western USA invasive ranges, as well as the native range. This haplotype was by far the most abundant and widespread of all the haplotypes recovered; individuals exhibiting it were collected in 21 rivers, across 13 states, and in both the USA and Mexico. The second most common haplotype, haplotype #45 (Fig. 2), was found in 8.75 % of individuals ($n = 42$) from 10 different rivers across 7 states.

Across the native range, the western lineage (Fig. 2) exhibited the highest haplotype diversity ($H_d = 0.981$, $SD = 0.031$), followed by the lineage encompassing the mid-western USA ($H_d = 0.976$, $SD = 0.004$), and the lineage with a south-western geographical footprint ($H_d = 0.954$, $SD = 0.017$). Despite encompassing two lineages, the invasive western USA populations exhibited an

average haplotype diversity comparatively lower than invasive eastern USA populations ($H_d = 0.916$, $SD = 0.014$; $H_d = 0.951$, $SD = 0.014$). Maximum haplotype diversity scores of 1 were recovered in seven different populations across the native range (Supplementary Table 1), while maximum haplotype diversity scores were only recovered in two invasive populations: Terrapin Creek (AL), and Dead River (AL; Supplementary Table 1). The native population with the lowest haplotype diversity was in Boyer River (IA), while the invasive population with the lowest haplotype diversity was in Virgin River (NV). Overall, haplotype diversity was not significantly different between native range ($H_d = 0.963$, $SD = 0.019$) and non-native range populations ($H_d = 0.901$, $SD = 0.020$; $p = 0.137$).

Other measures suggest that invasive populations harbor greater levels of genetic diversity than do native range populations. Notably, the average number of nucleotide differences in invasive populations ($k = 11.910$ bp, $SD = 5.445$) was over a third larger than that of native range populations ($k = 7.823$ bp, $SD = 3.233$; Supplementary Table 1). Higher nucleotide diversity was also found in introduced populations ($\pi = 0.080$, $SD = 0.044$) compared to native range populations ($\pi = 0.014$, $SD = 0.007$; Supplementary Table 1). However, some of these trends did not

hold when comparisons were drawn between individual rivers and sample sites. For example, invasive range samples from the Virgin River (NV) exhibited relatively low measures of genetic diversity ($k = 0.142$ bp, $\pi = 0.000$), while native range samples from the Canadian River (NM) exhibited comparably high levels of genetic diversity ($k = 7.667$ bp, $\pi = 0.010$).

Genetic structure

AMOVA comparisons revealed that most of the observed genetic variation across our sampled populations reflects a combination of local and regional heterogeneity. Most of the genetic variation among the native range populations from the mid-western lineage was attributable to differences among drainages (64.29 %, $\Phi_{CT} = 0.642$, $p = 0.000$), although genetic differentiation among populations within drainages and within populations were also highly significant (21.99 %, $\Phi_{SC} = 0.616$, $p = 0.000$ and 13.72 %, $\Phi_{ST} = 0.862$, $p = 0.000$, respectively). Comparable results were found when excluding sites represented by only one individual. Native populations from the south-western lineage did not show significant differentiation among drainages, but harbored a higher percentage of genetic variation among populations within drainages (95.91 %, $\Phi_{SC} = 0.963$, $p = 0.000$) and a lower percentage within populations (3.69 %, $\Phi_{ST} = 0.963$, $p = 0.000$) than in the mid-west. Neither the western lineage nor the Brazos lineage showed significant genetic structure at any level. In the invasive ranges, the extent of variation attributable to differences among collection sites differed between the eastern and western USA. A large portion of genetic variation in invaded western USA drainages reflects heterogeneity among collection sites ($\Phi_{ST} = 0.409$, $p < 0.001$), whereas variation in invaded eastern USA drainages does not reflect significant differences among collection sites (6.11 %, $\Phi_{ST} = 0.061$, $p = 0.099$). Comparing invaded eastern and western USA ranges explained the largest amount of variation across invasive populations (75.54 %, $\Phi_{CT} = 0.755$, $p < 0.001$). Overall, genetic differentiation between native and invasive populations, either considered altogether, or separately by western and eastern USA drainages, was non-significant (-6.35 %, $\Phi_{CT} = -0.063$, $p = 0.672$ and -14.48 %, $\Phi_{CT} = -0.145$, $p = 0.897$, respectively). However, considering drainages separately for each lineage showed significant genetic differentiation between native and invasive populations (36.04 %, $\Phi_{CT} = 0.360$, $p = 0.001$ for the mid-western lineage and 24.64 %, $\Phi_{CT} = 0.246$, $p = 0.030$ for the west lineage).

Invasion scenarios

The PCA pre-evaluation showed that the observed data fall within the cluster of simulated data points based on all

invasion scenarios, indicating that comparative analyses are appropriate to pursue (Supplementary Fig. 2). ABC analyses support an invasion scenario with geographically and temporally independent introductions into the eastern and western USA. The best supported model was Scenario 5 (PP = 0.6901, CI = 0.6355–0.7446; Supplementary Fig. 3), which proposes the existence of an ancestral admixed population with co-occurring haplotypes from the mid-western and western lineages from which the non-native range in the western USA was founded, and an independent invasion exclusively from mid-western lineage populations into the non-native range in the eastern USA. The second and third best supported scenarios were Scenario 4 and 3, which showed much lower relative posterior probabilities and no overlapping confidence intervals with Scenario 5 (PP = 0.2032, CI = 0.1544–0.2521, and PP = 0.1065, CI = 0.0798–0.1332). The probability of rejecting Scenario 5 when true (i.e., Type I error) was 0.564, and the probabilities of choosing it when Scenarios 3 and 4 were true (i.e., Type II errors) were 0.274 and 0.258. In the invaded ranges, the scenarios proposing one introduction event followed by within-range expansions (Scenarios W1 and E1) showed the highest support (W1: PP = 0.9309, CI = 0.9236–0.9382, and E1: PP = 0.6639, CI = 0.6553–0.6725; Supplementary Fig. 4). In all cases the confidence intervals for the relative posterior probabilities of the best-fit models did not overlap with the other models tested. Type I errors for Scenarios W1 and E1 were 0.276 and 0.308, and Type II errors were 0.176 and 0.208.

Discussion

Biological invasions generally follow a pattern of introduction, establishment, and range expansion (Kolar and Lodge 2001), but introduced species often exhibit fragmented distributions within invasive ranges, possibly as the result of multiple independent introductions from the native range or as the result of stochastic secondary spread. Though prior studies documenting red shiner introductions have provided observational records of first accounts and localized dispersal within individual river systems (Page and Smith 1970; Moyle 2002; Walters et al. 2008), the records offer limited information on the origin and nature of the disjunct distribution of non-native populations across North America. Through the analysis of mtDNA sequence variation, we reconstructed genealogical relationships among native and non-native populations to understand the history of red shiner invasions and to clarify the potential for future spread. ABC analyses suggest that introduced populations of red shiner across the eastern and western USA resulted from multiple independent introductions from the native range, which is consistent with prior

suggestions that introductions have largely been the result of unrelated bait bucket releases that have given rise to regionalized spread (Fuller et al. 1999; Walters et al. 2008). However, we also found that a highly divergent but phenotypically cryptic evolutionary lineage (the western clade, which likely warrants species recognition) has been introduced to the western USA and that multiple evolutionary lineages co-occur within the Virgin River (AZ). Thus estimates of potential spread under current and future climate conditions (Poulos et al. 2012; Poulos and Chernoff 2014) should be revisited to account for phylogenetic diversity within and among invasive populations. Management and monitoring of invasive populations also should account for the possibility of post-introduction hybridization among distinct evolutionary lineages in addition to hybridization with native congeners.

Phylogenetic diversity and genetic variation of native range red shiner

The taxonomy and phylogeny of *C. lutrensis* are not well resolved; since the species was originally described, different forms of “*lutrensis*” have been distinguished and more than a dozen names have been given to local variants across the native range (Contreras-Balderas 1975; Matthews 1987; Mayden 1989; Broughton and Gold 2000; Schönhuth and Mayden 2010). Presently, *C. lutrensis* is considered to be a widely distributed, heterogeneous species complex that consists of at least four monophyletic groups occurring in four distinct geographic regions (Richardson and Gold 1995; Schönhuth and Mayden 2010). Two of the four monophyletic groups that occur in the southern USA and Mexico have been considered to be different species (*C. forlonensis*, in central-east Mexico; *C. sp. 1* in northern Mexico-southern USA). Consistent with Osborne et al. (2015), we discovered an additional, previously undescribed and well supported western lineage from samples collected in north-central Texas, south-west Arkansas, and north-central Mexico, that is strongly divergent compared to other native range red shiner lineages. For example, approximately 11 % sequence divergence differentiates the newly detected lineage from the mid-western lineage. Assuming a mtDNA sequence divergence rate of approximately 1 % per million years (Estabrook et al. 2007; Smith and Dowling 2008; Hollingsworth et al. 2013), these two lineages diverged approximately 11 million years ago.

Like in many other North American fishes, the phylogeographic structure of red shiner lineages appears to reflect Pleistocene glaciation (Bermingham and Avise 1986; Richardson and Gold 1995; Soltis et al. 2006; Blum et al. 2008). Prior studies suggest that the evolutionary lineages ascribed to *C. lutrensis* arose well before the onset

of Pleistocene glaciation, and support the hypothesis that one or more lineages entered the region directly affected by glaciation during or after the Pleistocene (Richardson and Gold 1995). Similar to prior findings (Richardson and Gold 1995), our phylogenetic analysis of *C. lutrensis* within the native range revealed that the mid-western lineage exhibits the largest number of haplotypes and geographical extent within the species complex. However, our findings indicate that genetic variation within the mid-western and southwestern lineages largely reflects local scale heterogeneity, and that significant genetic structure occurs among drainages within the mid-western clade. This pattern of genetic structure could be the result of range expansion by means of stream capture and interconnecting drainages, following isolation in southern refugia during the last glacial maximum (Conner and Sutkus 1986; Richardson and Gold 1995; Schönhuth et al. 2008, 2011). The southwestern lineage, on the other hand, appears to exhibit a more restricted distribution despite encompassing six closely related clades with at least five named species that are likely of Pleistocene origin (Schönhuth and Mayden 2010). The high degree of heterogeneity and species co-occurrence in these drainages suggests that limited range expansion and secondary contact occurred following divergence in allopatry due to geographical isolation and divergence during the last glaciation. Conversely, we did not recover any significant genetic structure within the western or the Brazos lineages. In both cases, a pattern of geographically restricted lineages with low genetic diversity was observed, which could be attributable to low dispersal and long term isolation (Conner and Sutkus 1986; Richardson and Gold 1995).

Phylogenetic diversity and invasion history of non-native red shiner

Genetic analysis of invasive species can provide valuable information on demographic and evolutionary processes contributing to invasion success as well as colonization pathways and source populations (Darling and Blum 2007; Muirhead et al. 2008; Estoup and Guillemaud 2010; Guillemaud et al. 2010). ABC analyses provide an additional line of supporting evidence suggesting that the invasion of red shiner across the eastern USA is distinct from the invasion of the western USA. Hypothesized scenarios proposing single introductions of *C. lutrensis* from the native range into the western invasive range, and from the west to the eastern USA, were readily ruled out. Relatively large probabilities of Type I error complicated selection of the prevailing scenario, reflecting low confidence in distinguishing it from Scenarios 3 and 4. Nonetheless, the highest and non-overlapping confidence intervals of Scenario 5 posterior probability (over

Scenarios 3 and 4) together with much smaller Type II errors, elevate confidence in identifying Scenario 5 as the prevailing scenario (Bertorelle et al. 2010; Bermond et al. 2012).

Our findings indicate that the western USA introduction originated from an admixed population including individuals from the mid-west and western lineages in the native range, while eastern USA introductions derive exclusively from the mid-west lineage. Only haplotype #12, the most common haplotype ($n = 109$) was found in both the western and eastern invaded ranges. Haplotypes contributing to the western invasions mostly belonged to rivers harboring the western lineage, and matched native range haplotypes found in the Canadian and Pecos River (NM), Wichita River (TX), and San Pedro River (Mexico), and one mid-western haplotype found in the Kaskaskia River, (IL). The eastern introductions, however, represent a wider geographical sample from the native range with shared haplotypes from five states and Mexico. The repeated patterns of shared haplotypes provide some indication of potential native sources of introduction including the Arkansas River (AR) and the Salt River (MO), each of which shared the same two haplotypes with five non-native populations in North Carolina, Alabama and Georgia. Similarly, two haplotypes from the Verdigris River (KS) were also found in four populations in Alabama and Georgia. ABC models suggest, however, that secondary regionalized spread occurred within both the eastern and western invasive ranges following initial introduction events. This indicates either that secondary introductions from sources within the invasive ranges have occurred (i.e., via local transplantation) or that there is regional connectivity among drainages, possibly reflecting long-distance dispersal. This finding contrasts with some prior studies (Walters et al. 2008) suggesting that regional spread, even within a specific basin (e.g., the upper Mobile basin), is possibly an outcome of multiple independent introductions from the native range to geographically proximate basins.

Evidence of phylogenetic diversity clarifies long-standing questions about the identity and origins of red shiner introductions to the western USA (Moyle 1976; Jennings and Saiki 1990). We recovered two evolutionary lineages of *Cyprinella* within the Virgin River (AZ), including one that has not been previously described. The presence of two separate evolutionary lineages (likely warranting species recognition) of *Cyprinella* in the Virgin River highlights the possibility that putative red shiner introductions (i.e., introductions of fish that exhibit comparable phenotypes) across other regions of the USA may encompass cryptic invasions of multiple *Cyprinella* lineages or species. Given the distribution of the two lineages within the native range, it is possible that co-occurrence in the Virgin River is an outcome of two or more transplant events. Yet, ABC

modeling suggests that a multi-species “pool” of *Cyprinella* were introduced to this area in a single introduction event. This scenario is consistent with the history of aquaculture and use of red shiner as a bait fish as well as commercial sale of red shiner in the aquarium trade. The presence of the undescribed western lineage of *Cyprinella* in the San Joaquin River (CA) further illustrates that red shiner invasions are a mosaic of phylogenetic diversity. Our findings affirm that a genetically divergent lineage of *Cyprinella* was introduced to the San Joaquin River, which is consistent with evidence of phenotypic differences noted by Jennings and Saiki (1990), who questioned whether fishes introduced to California were red shiner or a closely related congener like *C. suavis* (Schönhuth and Mayden 2010).

The success of aquatic invasive species has often occurred despite what some have labeled a ‘genetic paradox’ (Sakai et al. 2001; Allendorf and Lundquist 2003; Roman and Darling 2007). Small founding populations of introduced species are expected to have genetic diversity that is lower than that of native populations as a result of demographic bottlenecks (Allendorf and Lundquist 2003). Small population size increases the risk of extinction and reduced genetic variation due to founder effects is expected to limit the ability of a newly established population to respond to novel conditions (Frankham 2005). Yet many introduced species either overcome or simply do not experience the challenge of low genetic diversity (Roman and Darling 2007). Some invasive populations, for example, do not exhibit comparably low genetic diversity as a consequence of successive introductions from a single or multiple native population (e.g., Blum et al. 2007). Admixture and interspecific hybridization can also promote invasion success by elevating genetic diversity, enabling introgression of locally adapted functional traits from native congeners, or by stimulating post-invasion evolution of novel genotypes (Ellstrand and Schierenbeck 2000; Lee 2002). Our findings further illustrate that invasive species are not necessarily constrained by low genetic diversity. Though local populations of non-native red shiner exhibited comparably lower levels of haplotype diversity, overall levels of diversity were higher in the invasive rather than native range. This is likely an outcome of multiple independent introductions as well as transplanted populations originating from mixed source pools. The occurrence of multiple evolutionary lineages in the Virgin River also raises the possibility of post-introduction hybridization between naturally allopatric evolutionary lineages. Though hybridization between red shiner and native congeners is a well-recognized threat to native biodiversity (Walters et al. 2008; Blum et al. 2010; Ward et al. 2012; Glotzbecker et al. 2015), it is possible that cryptic hybridization could be an additional threat. Invasions could be driven by

hybrids of non-native evolutionary lineages (e.g., Lack et al. 2012), or novel outcomes of reticulate hybridization between native congeners and multiple non-native evolutionary lineages or hybrids thereof (McDonald et al. 2008).

Disjunct distributions and forecasting spread

Understanding the origins and potential interactions between disjunct populations of invasive species can enable more accurate forecasting of future spread, particularly when models are intended to reflect physiological responses to changing climate conditions (Buckley et al. 2011; Araújo and Peterson 2012; Poulos et al. 2012). Forecasts of red shiner distributions under contemporary and future climate conditions (Poulos et al. 2012; Poulos and Chernoff 2014) are based on the dual premise that red shiner are a single evolutionary lineage that exhibits broad environmental tolerances and that red shiner are well adapted to succeed in novel, harsh or demanding conditions (Matthews and Hill 1977; Matthews 1985; Yu and Peters 2002; Walters et al. 2008). Our findings suggest that models forecasting spread of red shiner could be improved by accounting for the possibility of physiological or niche variation across the breadth of phylogenetic diversity that occurs in native and non-native populations (e.g., Compton et al. 2010; Kelley et al. 2011). This could afford more accurate perspectives on the likelihood of range contraction, expansion, or shifts across regions occupied by different evolutionary lineages. Additionally, predictive models likely could be improved if based on the understanding that some introduced populations encompass multiple evolutionary lineages and that regional invasions have likely proceeded via secondary spread.

The introduction of invasive red shiner across the USA serves as an example of how disjunct distributions can originate from multiple, independent introductions from a complex of phylogenetically diverse source populations (e.g., Blum et al. 2007; Darling et al. 2008). Our findings suggest that the disjunct distribution of red shiner across North America is a collection of regional invasions with distinct origins rather than the stepwise advance of one or more unified invasion front. Thus localized control or removal may be effective in managing non-native red shiner, including further spread to areas of conservation concern. The risk of continued spread could be better managed, and perhaps reduced, if forecasting models account for the distribution of evolutionary lineages that may occupy distinct fundamental niches. Further monitoring and genetic assessments of non-native populations could also help define invasion risk, particularly in the western USA where co-occurring non-native evolutionary lineages could hybridize.

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Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

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